

Nitin Jeyakumar

Work Authorization: US Citizen

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SUMMARY

Results-oriented Software Engineer specialized in developing robust web applications and integrating AI-powered technologies, improving system performance by 25% and query efficiency by 40%. Proficient in modern tech stacks, including React.js, Node.js, AWS, and PostgreSQL, with hands-on experience in database optimization and cloud-based architectures. Skilled in Agile methodologies, CI/CD pipelines, and microservices, achieving a 20% boost in workflow efficiency. Dedicated to crafting innovative solutions that address complex challenges and drive user satisfaction.

EDUCATION

University of Illinois at Chicago

Chicago, IL

Master of Science in Information Systems, GPA: 3.90/4.00 (graduating in May 2025)

August 2023-May 2025

Relevant Coursework: Application Development, Advanced Database Management Systems, Data Mining, Introduction to Artificial Intelligence, Machine Learning, Cloud Computing and Networks.

Coimbatore Institute of Technology

TN, India

Bachelor of Science in Computer Science, GPA: 3.74/4.00

July 2019-July 2023

Relevant Coursework: Programming in Java, Data Structures & Algorithms, Object-Oriented Programming, Mobile App Development, Computer Architecture, Operating Systems, Compiler Design, Advanced Algorithms.

SKILLS

Programming Languages & Tools: Python, JavaScript, Java, C++, React.js, Node.js, CI/CD, Postman, SQL, PostgreSQL.

Cloud Platforms and Services: AWS (EC2, S3, DynamoDB, Lambda, RDS, CloudFormation, API Gateway, CloudWatch, Glue, Kinesis, IAM, OpenSearch/Elasticsearch, Bedrock)

Machine Learning & AI: TensorFlow, PyTorch, Transformers, Natural Language Processing (NLP), Machine Learning (SVM, CNN, RAG, Hyperparameter Tuning, Reinforcement Learning), Generative AI, Large Language Models (LLMs).

Libraries & Frameworks: Pandas, NumPy, MongoDB, REST APIs, GraphQL, Docker, Langchain.

WORK EXPERIENCE

Headstarter

Remote, US

Software Engineer Fellowship

July 2024-September 2024

- Led a team of 4 engineering fellows in designing scalable projects using **MVC design patterns** and **Agile methodologies**.
- Built production-ready applications and APIs with Next.js, OpenAI, Pinecone, and Stripe, achieving 98% accuracy.
- Streamlined workflows with **CI/CD pipelines, Git, and microservices**, boosting efficiency by 20%.

Anna University

TN, India

Full Stack Developer

May 2022-July 2023

- Developed and optimized the university web application using React.js, reducing load times by 15% and improving usability.
- Designed and integrated **RESTful APIs** to enhance dataflow between services, reducing data retrieval times by 10%.
- Implemented **NoSQL database solutions**, increasing backend scalability and supporting 5,000+ data records.
- Strengthened application efficiency by 10% through end-to-end solutions using **MERN stack**.

ACADEMIC PROJECTS

AI-Powered Document Intelligence with AWS Bedrock:

- Engineered a document retrieval system using **FAISS** to manage over **10,000 embeddings**, enabling efficient search and retrieval.
- Integrated **Amazon Bedrock** AI services, increasing response accuracy by 25% based on user feedback and testing.
- Enhanced system scalability, reducing query response time by 40% through backend optimization and parallel processing.
- Demonstrated proficiency in **NLP techniques** such as **tokenization and embedding generation** and utilized cloud-based AI tools for seamless integration.

Driver Drowsiness Alerting System

- Developed a drowsiness alerting system using **Python, OpenCV** and other **deep learning techniques**.
- Implemented machine learning techniques (**SVM algorithm and CNN algorithm**) for real-time facial feature detection.
- Elevated processing speed by 25%, slashing false positives by 15% in the Driver Drowsiness Alerting System through targeted algorithmic enhancement.

Tomato (Food Delivery Application)

- Developed a real time food delivery application using **React.js** for frontend and **Node.js** with **Express.js** for the backend.
- Utilized **MongoDB** for data management and integrated secure payment processing using Stripe.

Certifications

AWS Certified Solutions Architect - Associate